

Comprehensive Test Plan

1. Objective

The objective of this test plan is to verify that all application functionalities work according to business requirements and provide a reliable, secure, and user-friendly experience.

Testing aims to:

- Identify defects before production release
 - Validate functional requirements
 - Verify API integrations
 - Ensure security compliance
 - Evaluate performance under load
 - Confirm stability after fixes
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2. Scope of Testing

In Scope

Functional Testing

- User Registration
- Login Authentication
- Dashboard Operations
- Data Upload
- Data Processing
- Report Generation
- Search Features
- Notifications

API Testing

- GET Requests
- POST Requests
- PUT Requests
- DELETE Requests
- Authentication APIs

Security Testing

- Password Protection
- Session Management
- Authorization Checks

Performance Testing

- Response Time
- Concurrent Users
- Load Handling

Regression Testing

- Verification of fixed defects
 - Existing feature validation
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Out of Scope

- Third-party vendor systems
 - Hardware testing
 - Mobile application testing (if unavailable)
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3. Test Strategy

Functional Testing

Approach

Validate all business workflows against requirements.

Example

User Login

Input:

username: testuser
password: Test@123

Expected:

User redirected to dashboard

API Testing

Tool

- Postman

Validation

- Status Codes
- Response Time
- Response Body
- Data Integrity

Example:

GET /users

Expected:

200 OK

Security Testing

Validation Areas

- SQL Injection
- XSS
- Authentication
- Authorization

Example:

Input:

' OR 1=1 --

Expected:

Login blocked

Performance Testing

Tool

- JMeter

Metrics

- Average Response Time
- Throughput
- Error Rate

Target:

Response Time < 3 sec

Regression Testing

Re-run all critical test cases after bug fixes.

Modules:

- Login
 - Dashboard
 - Upload
 - Reports
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4. Test Environment

Component	Configuration
OS	Windows 11
Browser	Chrome Latest
API Tool	Postman
Performance Tool	JMeter
Database	MySQL
Network	Stable Internet

5. Entry Criteria

Testing starts when:

- Requirements available
 - Application deployed
 - Test environment ready
 - Test data prepared
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6. Exit Criteria

Testing ends when:

- All planned tests executed
 - No Critical defects open
 - No High Severity defects pending
 - Test Summary Report approved
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7. Test Deliverables

Documents

- Test Plan
 - Test Cases
 - Test Execution Report
 - Defect Report
 - API Test Report
 - Regression Report
 - Test Summary Report
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8. Test Data Requirements

Valid Data

Field	Value
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Username	testuser
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Password	Test@123
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Invalid Data

Field	Value
Username	blank
Password	blank

9. Risk Assessment

Risk	Impact	Mitigation
Environment Downtime	High	Backup Environment
Requirement Changes	Medium	Requirement Review
API Unavailability	High	Mock APIs
Time Constraints	Medium	Priority Testing

10. Defect Severity Levels

Severity	Description
Critical	Application Crash
High	Major Function Failure
Medium	Partial Function Failure
Low	UI Issues

11. Test Metrics

Track:

Execution Metrics

- Total Test Cases
- Executed
- Passed
- Failed
- Blocked

Defect Metrics

- Critical Defects
- High Defects
- Medium Defects
- Low Defects

Example:

Metric	Count
Total Test Cases	100
Passed	92
Failed	8
Pass Rate	92%

12. Roles and Responsibilities

Role	Responsibility
QA Tester	Execute Tests
QA Lead	Review Results
Developer	Fix Defects
Project Manager	Approve Release

13. Test Execution Schedule

Phase	Duration
Test Planning	1 Day
Test Case Creation	2 Days
Functional Testing	3 Days
API Testing	2 Days
Security Testing	1 Day
Regression Testing	2 Days
Reporting	1 Day